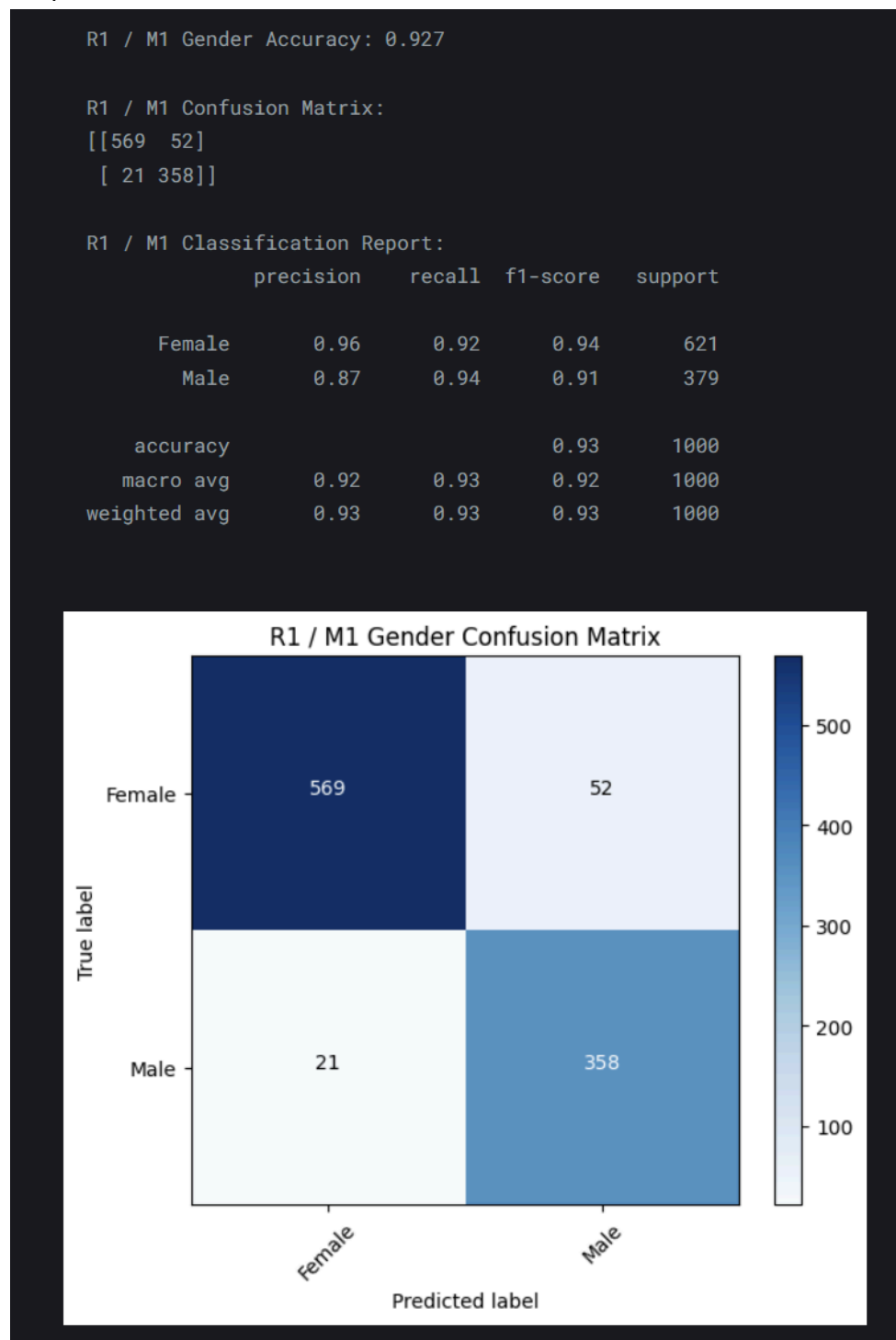


R1 - Classification accuracy and confusion matrix that is produced by deploying my model on the partition test



R2 - Classification accuracy and confusion matrix that is produced by deploying my model on the partition test

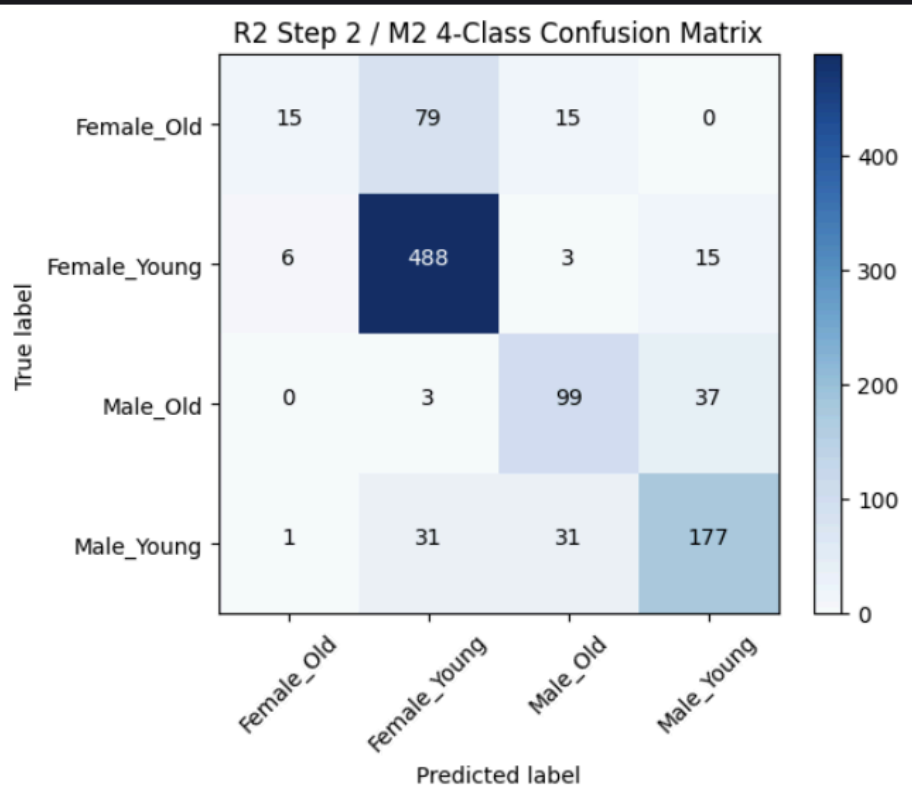
R2 Step 2 / M2 4-Class Accuracy: 0.779

R2 Step 2 / M2 4-Class Confusion Matrix:

```
[[ 15  79  15   0]
 [   6 488   3  15]
 [   0   3  99  37]
 [   1  31  31 177]]
```

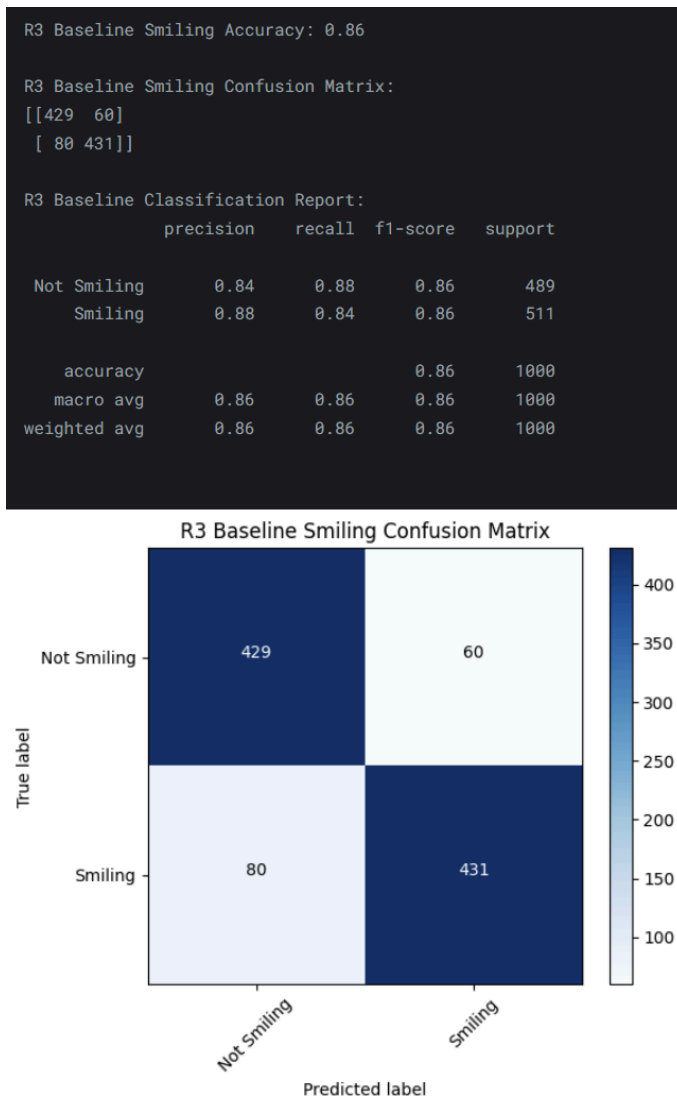
R2 Step 2 / M2 4-Class Classification Report:

	precision	recall	f1-score	support
Female_Old	0.68	0.14	0.23	109
Female_Young	0.81	0.95	0.88	512
Male_Old	0.67	0.71	0.69	139
Male_Young	0.77	0.74	0.75	240
accuracy			0.78	1000
macro avg	0.73	0.64	0.64	1000
weighted avg	0.77	0.78	0.75	1000

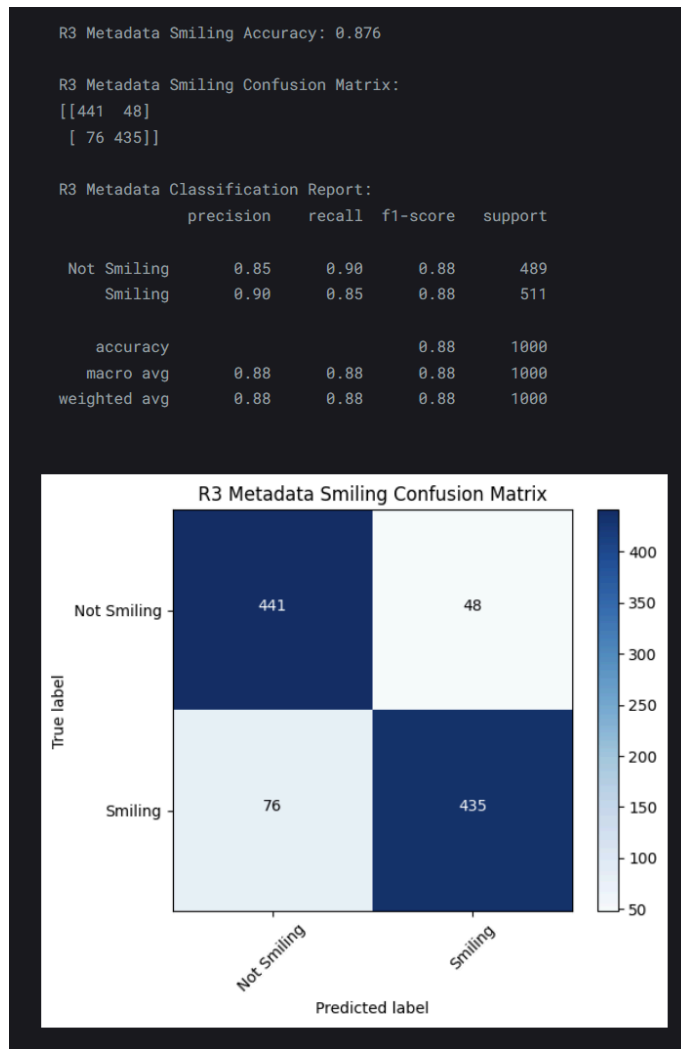


R3

A. This is the confusion matrix for the smiling target.



B. This is the classification accuracy and confusion matrix for the mouth width as a



predictor.

C. Looking at my results, they show that the baseline smiling accuracy was 0.86 or 86% and then the smiling accuracy with the mouth quartile was 0.876 or 87.5%. This shows that the model that incorporates the mouth_wide quartile got better accuracy than the baseline model. This leads to the conclusion that the mouth_wide predictor is useful metadata for smiling classification. While the increase in accuracy was pretty small, only 1.6%, meaning that the added data was useful at least a little bit. The increase was small because the model was already pretty accurate without the added data. The metadata was helpful since the mouth width is directly related to facial expression, it provides more information from more than just pixel data.